

# PAGASA SYNOP Validator

Operational Checking Ruleset  
Version 0.15.0 - Reviewed Branch Update

| Document status | Operational review guide                                          |
|-----------------|-------------------------------------------------------------------|
| Coding form     | WMO FM 12 SYNOP (AAXX), with PAGASA practices                     |
| Validator input | One complete observation at a time                                |
| Published build | Ruleset v0.15.0                                                   |
| Basis           | v0.14.3 operational rules plus reviewed collaborator improvements |

## Purpose

This document describes the checks implemented in the PAGASA SYNOP Validator for operational review, regression testing, and controlled maintenance. It does not replace the official WMO Manual on Codes, PAGASA observing instructions, or observer judgment.

**Important:** 'No coded errors detected' means only that the observation passed the rules currently implemented. It is not a certification that every observation or coding decision is correct.

# 1. Inputs, result levels, and bulletin schedule

| Input                 | Purpose                                                                               |
|-----------------------|---------------------------------------------------------------------------------------|
| Complete SYNOP code   | Required AAXX observation, preferably with bulletin heading and '=' end marker.       |
| MSLP 3 hours earlier  | Exact sign-category and ppp-magnitude check for 5appp.                                |
| MSLP 24 hours earlier | Exact signed check of Section 3 58/59 at 00 and 12 UTC.                               |
| Rainfall occurred     | Single operational indicator cross-checked with rainfall groups, iR, tR, and weather. |

## Finding levels

- **Error:** conflict with required syntax, schedule, arithmetic, or an implemented operational rule.
- **Warning:** possible code that needs observational review or missing supporting information.
- **Information:** an optional section or check is unavailable.
- **OK:** no implemented error or warning was produced.

## Bulletin schedule

- SMPH20 is expected at 00, 06, 12, and 18 UTC.
- SIPH20 is expected at 03, 09, 15, and 21 UTC.
- When present, bulletin YYGG is compared with AAXX YYGGiw.

## 2. Structure, syntax, and duplicate groups

| Element               | Implemented check                                                           |
|-----------------------|-----------------------------------------------------------------------------|
| AAXX / YYGGiw / Iliii | Required positions; valid day, hour, iw, and five-digit station identifier. |
| Mandatory groups      | iRixhVV and Nddff immediately follow the station identifier.                |
| Sections              | 333 begins Section 3 and 555 begins Section 5.                              |
| Group format          | Five figures or permitted solidi; missing '=' produces a warning.           |
| Section 3 order       | Standard Section 3 groups are checked for indicator order.                  |

### Semantic duplicate detection

Single-occurrence Section 1 families are compared by their meaning, not only by exact text. Thus **10313 10312** is an error because both are 1snTTT air-temperature groups. The same approach applies to dew point/humidity, station pressure, MSLP, pressure tendency, present/past weather, and the main cloud group. Legitimately repeatable groups remain allowed.

### 3. Pressure and local history

| Rule               | Check                                                                          |
|--------------------|--------------------------------------------------------------------------------|
| 4PPPP plausibility | Configured operational range: 870.0 to 1085.0 hPa.                             |
| 5appp magnitude    | Absolute current-minus-3-hour difference equals ppp exactly to 0.1 hPa.        |
| 5appp sign         | Positive: a=0-3; negative: a=5-8; zero: a=0,4,5.                               |
| 24-hour change     | At 00/12 UTC, 58 is positive/zero and 59 is negative; exact 0.1 hPa agreement. |
| Zero change        | 59000 is rejected; zero uses 58000.                                            |

#### Browser-local MSLP history

- A single structurally valid and plausible 4PPPP value is retained for 36 hours.
- Matching uses station identifier and SYNOP day/hour; manual entries take priority.
- Month-boundary matching is skipped when month/year cannot be known from YYGGiw.
- History stays only in the current browser/device and is not synchronized.

If previous pressure is unavailable, the validator requests it with a warning and does not create a false mismatch.

## 4. Rainfall, weather, and visibility

| Topic           | Implemented rule                                                                                 |
|-----------------|--------------------------------------------------------------------------------------------------|
| iR and location | Checks required presence and placement of 6RRRtR in Sections 1 and/or 3.                         |
| tR schedule     | tR=4 at 00; tR=1 at 06/12/18; tR=7 at 03/09/15/21 UTC.                                           |
| Autofill        | Uses the shared rainfall-evidence rule; 6000t, 6////, and snow-only evidence do not auto-select. |
| Manual choice   | A manually selected or cleared checkbox is preserved while editing.                              |
| Weather         | Cross-checks rainfall against applicable ww and W1/W2 indications.                               |

### Visibility rules

| Phenomenon             | Horizontal visibility treatment                                    |
|------------------------|--------------------------------------------------------------------|
| Light precipitation    | Suggested 9 km or more; lower value gives a warning.               |
| Moderate precipitation | 2 km to less than 9 km.                                            |
| Heavy precipitation    | 2 km or less; a value above 2 km is an error.                      |
| Present weather 40     | 2 km or less; a value above 2 km is an error.                      |
| Present weather 04-06  | No mandatory upper limit. Haze-intensity ranges are guidance only. |

## 5. Cloud rules and reportable layers

Cloud checks combine Nddff, h in iRixhVV, 8NhCLCMCH, and Section 3 8NsChshs groups.

### PAGASA 1-3-5 selection

- Report the lowest layer at 1 okta or more.
- Report the next higher layer at 3 oktas or more.
- Report the next higher layer at 5 oktas or more.
- Always report Cumulonimbus when not already selected; maximum four reportable groups.
- Order reportable groups from lower to higher bases.

### Independent layer amounts

Each Ns is estimated as if other layers did not exist, so layers can overlap. The validator does not add Section 3 Ns values to derive Nh. It does check that an individual low-cloud Ns does not exceed Nh. CB does not imply that no other low cloud exists.

Example: **84908 333 81915 83820 87280 94945** is consistent with Nh=4, one okta CB and three oktas Cumulus.

### Sky state

| N | Treatment                                                  |
|---|------------------------------------------------------------|
| 0 | Omit main and individual cloud groups.                     |
| 9 | Omit main cloud group; report 89/hshs vertical visibility. |
| / | Omit main and individual cloud groups.                     |

## 6. CB, gustiness, and monthly rainfall

### 949CD nature codes

| C | Meaning                           |
|---|-----------------------------------|
| 4 | Isolated Cumulonimbus             |
| 5 | Numerous Cumulonimbus             |
| 6 | Isolated Cumulus and Cumulonimbus |
| 7 | Numerous Cumulus and Cumulonimbus |

D is direction 0-9. Multiple 949CD groups may be used for CB in different directions. A valid example is **94968**: isolated Cumulus and Cumulonimbus to the north.

### Gustiness

A 904tt or 907tt group must be followed immediately by 911ff and 915dd. Orphan speed or direction groups are reported.

### Monthly rainfall

At 00 UTC on day 1, Section 5 may contain the six-hour rainfall group and the previous-month total. When the monthly value exceeds 1000 mm, the extended form is **6//// RRRRR**. The five-digit total is interpreted in tenths of a millimetre.

## 7. Release safeguards and limitations

### Regression coverage

- Exact 3-hour and 24-hour pressure differences, including 0.1 hPa failures.
- Main/intermediate bulletin mismatches and bulletin/AAXX time consistency.
- Heavy precipitation and ww=40 at and above the 2 km boundary.
- Valid and invalid high-cloud obscuration cases.
- Combined Cu/CB 949 codes, unrestricted ww=04-06 visibility, and semantic duplicate groups.
- Rainfall autofill exclusions and preservation of manual choices.

### Known limitations

- The validator sees coded text, not the original instrument readings or sky observation.
- Local pressure history cannot infer month/year and is not shared between devices.
- Suggestions require observer review; the program does not rewrite or transmit an observation automatically.

### Maintenance workflow

- Edit routine values in `dist/js/ruleset-config.js`.
- Edit meteorological logic in `dist/js/ruleset.js` and interface behavior in `dist/js/app.js`.
- Add a correct and incorrect regression example for every rule change.
- Run: `node tests/ruleset-regression.test.mjs`
- Review the v0.15.0 branch before merging it into main.

### Operational status

This is an operational review build. Continue comparison with official WMO and PAGASA documentation and record every accepted correction in the changelog.